

Anxiety Disorders



What is Anxiety????

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- Anxiety (i.e., a feeling of fear that is disproportionate to the actual threat or situation) is an **abnormal reaction** to a perceived threat to one's physical or psychological well-being.
- Anxiety involves the perturbation of several different neural systems, causing two basic components:
 - **Mental features** (e.g., worry, fear, difficulty concentrating) &
 - **Physical symptoms** (e.g., racing heart, shortness of breath, trembling, pacing).

When anxiety become disease???

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Pathological anxiety:

- Out of proportion to the actual threat
- Lasts far beyond the presence of the threat
- It impair normal functioning
- The anxiety experienced by a person **is excessive for the situation** (in intensity or duration) or very distressing, to the point that it interferes with daily functioning.

Cont.....

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- Anxiety disorder is constellation of disorders in which **anxiety** and associated **symptoms** are irrational or experienced at a level of severity that **impairs functioning**.
- In anxiety disorders the most prominent features are **anxiety** and **avoidance** which are irrational or impair functioning.

Epidemiology

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- The prevalence of common mental illness in the general population was 21.6%
- Around 36.43% of patients with comorbid illness
- Common mental illnesses include anxiety and depression

Classification of anxiety disorder:

1. Generalized Anxiety Disorder (GAD)

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- Excessive, persistent, unreasonable anxiety about everything's
- Persistent symptoms most days for at least 6 months.
- Worry about a number of matters and is accompanied by at least 3 psychological or physiologic symptoms.
- Has gradual onset (average age of 21 yrs).
- Chronic course, with multiple spontaneous exacerbations &

Presentation of GAD

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Psychological and cognitive symptoms

- Excessive anxiety
- Worries that are difficult to control
- Feeling nervous
- Poor concentration or mind going blank

Physical symptoms

- ☐ Restlessness
- ☐ Fatigue
- ☐ Muscle tension
- ☐ Sleep disturbance
- ☐ Irritability

Impairment

- ☐ Social, occupational, or other important functional areas
- ☐ Poor coping abilities

2. Panic Disorder

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- Panic attacks, which are sudden **feelings of terror when there is no real danger.**
- Serious **panic attacks**, followed by at least 1 mth of persistent concern; there must be at **least four physical symptoms** in addition to psychological symptoms.
- Symptoms reach a peak within **10 min** and usually last no more than 20 or 30 min.
- Many pts eventually develop **agoraphobia**, which is avoidance of specific situations (e.g., crowded places, bridges) where they fear a panic attack might occur and Pts may become homebound.

Symptoms of Panic Attack

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Psychological symptoms

- Depersonalization
- De-realization(a feeling that one's surroundings are not real)
- Fear of losing control
- Fear of going crazy
- Fear of dying

Physical symptoms

- Abdominal distress
- Chest pain or discomfort
- Chills, Hot flushes
- Dizziness or lightheadedness
- Feeling of choking
- Palpitations, Nausea
- Paresthesias
- SOB, Sweating, Tachycardia
- Trembling or shaking

3. Social Anxiety Disorder (SAD)

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- The essential feature of SAD is an intense, irrational, and persistent fear of being **negatively evaluated** in a social or performance situation.
- Exposure to the feared situation → **Panic attack.**
- It interferes with daily routine or social /occupational functioning.
- It is a chronic disorder, onset in the **teens.**

Presentation of SAD

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Fears

- Being scrutinized by others
- Being embarrassed
- Being humiliated

Some feared situations

- Addressing a group of people
- Eating or writing in front of others
- Interacting with authority figures
- Speaking in public
- Talking with strangers
- Use of public toilets

Physical symptoms

- Blushing
- "Butterflies in the stomach"
- Diarrhea
- Sweating
- Tachycardia
- Trembling

Types

- Generalized type: fear and avoidance extend to a wide range of social situations
- Nongeneralized type: fear is limited to one or two situations

4. Post-traumatic Stress Disorder (PTSD)

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- In PTSD, exposure to a traumatic event causes immediate intense fear, helplessness, or horror.
- Pts must have
 - At **least One re-experiencing** symptom,
 - Three signs or symptoms of persistent **avoidance of stimuli**, and
 - At least two symptoms of **increased stimulation**.

Post-traumatic Stress Disorder

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- Symptoms present longer than 1 month & cause significant distress or impairment.
- Pts with PTSD have
 - Poor prognosis,
 - Concurrent depression or anxiety disorder.
 - Comorbid alcohol abuse or dependence (men),
 - Attempt suicide.

Presentation of PTSD

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Reexperiencing symptoms

- Recurrent, intrusive distressing memories of the trauma
- Recurrent, disturbing dreams of the event
- Feeling that the traumatic event is recurring (e.g., dissociative flashbacks)
- Physiologic reaction to reminders of the trauma

Avoidance symptoms

- Avoidance of conversations about the trauma
- Avoidance of thoughts or feelings about the trauma
- Avoidance of activities that are reminders of the event
- Avoidance of people or places that arouse recollections of the trauma
- Inability to recall an important aspect of the trauma
- Anhedonia
- Estrangement from others
- Restricted affect
- Sense of a foreshortened future (e.g., does not expect to have a career, marriage)

Hyperarousal symptoms

- Decreased concentration
- Easily startled
- Hypervigilance
- Insomnia
- Irritability or angry outbursts

Subtypes

- Acute: duration of symptoms is less than 3 months
- Chronic: symptoms last for longer than 3 months
- With delayed onset: onset of symptoms is at least 6 months posttrauma

Screening questions

- Have you ever experienced a significant trauma in your life?
- Did this experience have a lasting negative impact or change your life?

5. Obsessive Compulsive Disorder (OCD):

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- ▣ Obsessive or **disturbing thoughts** that cannot be controlled and that are **repetitive** behavior.
- ▣ Compulsions are **ritualistic behaviors** (e.g., washing the hands, combing the hair, cleaning the house, counting, checking)

Classification

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6. Phobic disorders (including social anxiety disorder):

- ▣ marked and persistent fear and anxiety in social or performance situations that are recognized as excessive or unreasonable.
- ▣ These situations are either avoided or endured with intense anxiety.

- The **DSM-IV-secondary anxiety** disorders include:
 - “anxiety disorder due to a general medical condition”
and
 - “substance-induced anxiety disorder”.

Pathophysiology

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□ **Noradrenergic model:**

- ✓ *Autonomic nervous system* of anxious pts is hypersensitive & overreacts to various stimuli.
- ✓ The **locus ceruleus** may have a role in **regulating anxiety**, as it activates NE release & stimulates the SNS
- ✓ **Chronic noradrenergic over activity** down regulates $\alpha 2$ -adrenoreceptors in pts with **GAD & PTSD**.
- ✓ Pts with SAD appear to have a **hyper responsive** adrenocortical response to psychological stress.

Cont.....

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□ ***GABA receptor model***

- ✓ GABA is the major inhibitory neurotransmitter in the CNS.
- ✓ Anxiety symptoms are linked underactivity of GABA systems or downregulated central **Benzodiazepine receptors**.

□ ***5-HT model***

- ✓ 5-HT is primarily an inhibitory neurotransmitter that is used by neurons originating in the **raphe nuclei** of the brain stem and projecting diffusely throughout the brain (e.g., cortex, amygdala, hippocampus, and limbic system).

Diagnosis

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□ Examination

- Complete physical and mental status;
- Appropriate laboratory tests; and
- Medical, psychiatric, and drug history.

□ Anxiety symptoms may be associated with **medical illnesses** or **drug therapy**,

□ May present in major psychiatric illnesses

Common Illnesses Associated with Anxiety Symptoms

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Cardiovascular

Angina, arrhythmias, congestive heart failure, ischemic heart disease, myocardial infarction

Endocrine and metabolic

Cushing's disease, hyperparathyroidism, hyperthyroidism, hypothyroidism, hypoglycemia, hyponatremia, hyperkalemia, pheochromocytoma, vitamin B₁₂ or folate deficiencies

Neurologic

Dementia, migraine, Parkinson's disease, seizures, stroke, neoplasms, poor pain control

Respiratory system

Asthma, chronic obstructive pulmonary disease, pulmonary embolism, pneumonia

Others

Anemias, systemic lupus erythematosus, vestibular dysfunction

Drugs Associated with Anxiety Symptoms

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- ❑ **Anticonvulsants: carbamazepine, phenytoin**
- ❑ **Antidepressants: selective serotonin reuptake inhibitors, bupropion, serotonin–norepinephrine reuptake inhibitors**
- ❑ **Antihypertensives: clonidine, felodipine**
- ❑ **Antibiotics: quinolones, isoniazid**
- ❑ **Bronchodilators: albuterol, theophylline**
- ❑ **Corticosteroids: prednisone**
- ❑ **Dopamine agonists: amantadine, levodopa**
- ❑ **Herbals: ginseng, ephedra**
- ❑ **Illicit substances: marijuana**
- ❑ **Nonsteroidal antiinflammatory drugs: ibuprofen, indomethacin**
- ❑ **Stimulants: amphetamines, methylphenidate, nicotine, caffeine, cocaine**
- ❑ **Sympathomimetics: pseudoephedrine, phenylephrine**
- ❑ **Thyroid hormones: levothyroxine**
- ❑ **Toxicity: anticholinergic, antihistamines, digoxin**

Treatment

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General Anxiety Disorder

□ Desired outcome

- ✓ Reduce severity, duration & frequency of symptoms
- ✓ Improve overall functions
- ✓ **long-term goal: minimal or no anxiety** or depressive symptoms, no functional impairment, and improved QOL.

Non-pharmacologic modalities

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- Short-term counseling, stress management, cognitive therapy, meditation, supportive therapy, and exercise.
- Avoid caffeine, stimulants, excessive alcohol & diet pills.
- Cognitive behavioral therapy (**CBT**) the most important for long term treatment.

Treatment: drug choices

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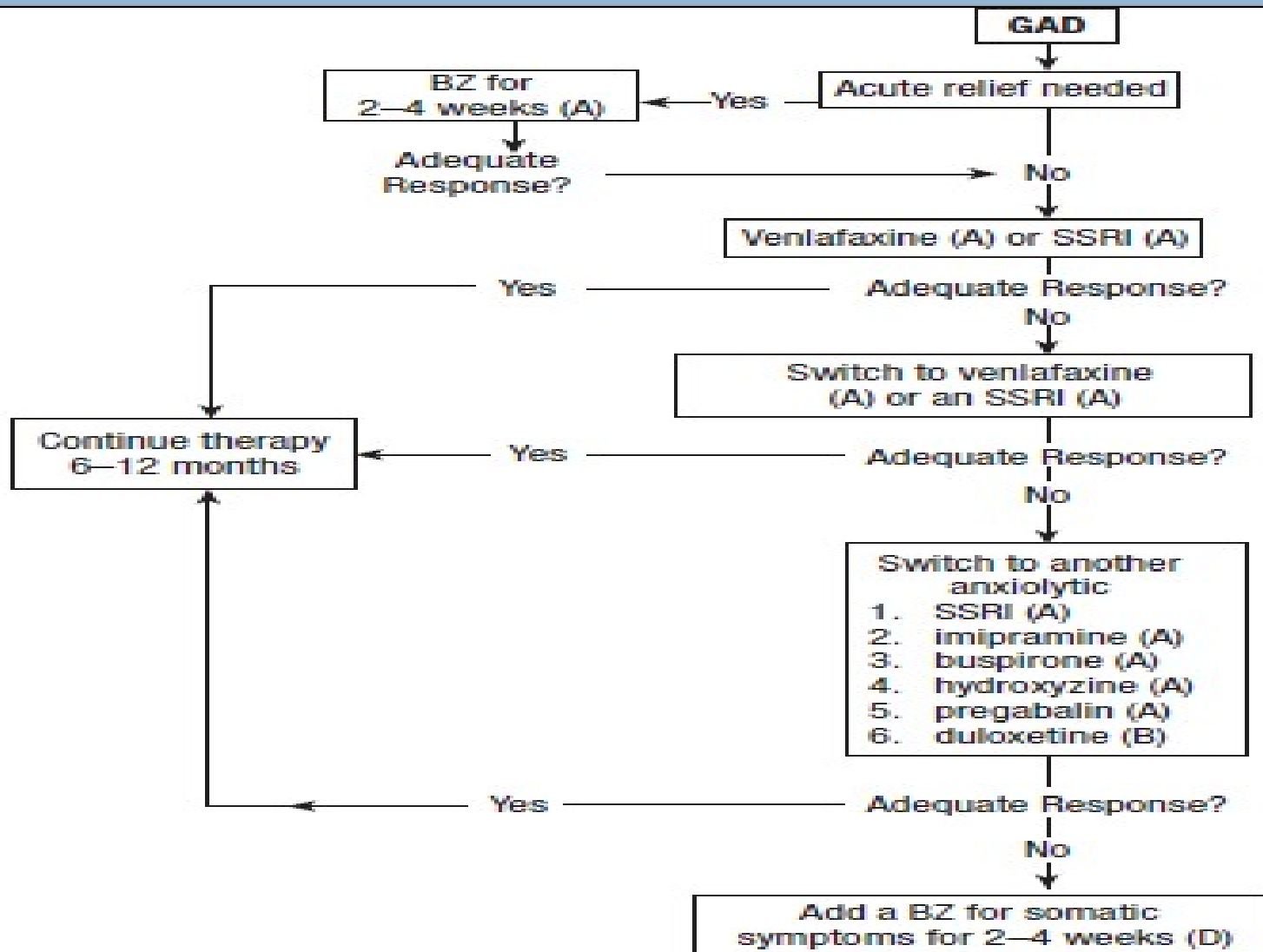
Anxiety Disorder	First-Line Drugs	Second-Line Drugs	Alternatives
Generalized anxiety	Duloxetine Escitalopram Paroxetine Venlafaxine XR	Benzodiazepines Buspirone Imipramine Sertraline	Hydroxyzine Pregabalin
Panic disorder	SSRIs Venlafaxine XR	Alprazolam Clomipramine Clonazepam Imipramine	Phenelzine
Social anxiety disorder	Escitalopram Fluvoxamine Paroxetine Sertraline Venlafaxine XR	Citalopram Clonazepam	Buspirone Gabapentin Mirtazapine Phenelzine Pregabalin

TABLE 66–8**Nonbenzodiazepine Antianxiety Agents for Generalized Anxiety Disorder**

Drug	Brand Name	Initial Dose	Usual Range (mg/day)*	Comments
<i>Antidepressants</i>				
Duloxetine	Cymbalta	30 or 60 mg/day	60–120	FDA-approved
Escitalopram	Lexapro	10 mg/day	10–20	FDA-approved, available generically
Imipramine	Tofranil	50 mg/day	75–200	Available generically
Paroxetine	Paxil Pexeva	20 mg/day	20–50	FDA-approved, available generically, avoid in pregnancy
Sertraline	Zoloft	50 mg/day	50–200	Available generically
Venlafaxine XR	Effexor XR	37.5 or 75 mg/day	75–225 ^b	FDA-approved, available generically
<i>Azapirone</i>				
Buspirone	BuSpar	7.5 mg twice daily	15–60 ^b	FDA-approved, available generically
<i>Diphenylmethane</i>				
Hydroxyzine	Vistaril	25 or 50 mg four times daily	200–400	FDA-approved, available generically, approved in children for anxiety and tension in divided daily doses of 50–100 mg
<i>Anticonvulsant</i>				
Pregabalin	Lyrica	50 mg three times daily	150–600	Dosage adjustment required in renal impairment
<i>Atypical antipsychotic</i>				
Quetiapine XR	Seroquel XR	50 mg at bedtime	150–300	

Treatment algorithm

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Antidepressants

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- Antidepressants have now surpassed benzodiazepines as the recommended first-line treatment for most anxiety disorder patients, even though benzodiazepines are still widely prescribed.
- For acute and long-term management of GAD.
- They are considered the **treatment of choice** for long-term management of chronic anxiety, especially in the presence of depressive symptoms.

Cont.....

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- **Anxiolytic effects of benzodiazepines occur almost immediately, whereas the effects of antidepressants occur gradually over several weeks.**
- For this reason, benzodiazepine therapy to be prescribed in combination with antidepressant during initial treatment of many anxiety disorders.

Cont.....

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- FDA approved drugs for GAD
 - **Venlafaxine XR, duloxetine, paroxetine and escitalopram**
- **Sertraline** is also effective.
- **Imipramine may be used when pts fail to respond to SSRIs.**

Benzodiazepine Therapy(anxiolytic agents)

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- ❑ Improvement occurs in the first 2 wks of therapy.
- ❑ BZs are more effective for **somatic** & autonomic symptoms of GAD, while antidepressants more effective for the **psychic symptoms** (worry & apprehension).

Side Effect of BZs

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□ **CNS depression**

- **Tolerance** usually develops to this effect.
- Disorientation, psychomotor impairment, confusion, aggression, excitement

□ **BZ abuse**

- **BZ dependence:** appearance of withdrawal symptoms upon abrupt discontinuation.

BZ discontinuation

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- If BZs are abruptly discontinued, three events can occur:
 - ✦ Rebound symptoms – immediately
 - ✦ Recurrence or relapse
 - ✦ Withdrawal – within 24-48 hr for short $t_{1/2}$
 - ✦ 3-8 days for long $t_{1/2}$

- A 25% per wk reduction in dosage until 50% of the dose is reached, then dosage reduction by 1/8th every 4 to 7 days.
- If therapy exceeds 8 wks, a taper over 2 to 3 wks is recommended, but if duration of treatment is 6 months, a taper over 4 to 8 wks should ensue.
- Longer durations of treatment may require a 2- to 4-mth **taper.**

- A BZ with a long $t_{1/2}$ (**diazepam, clonazepam**) may be substituted for a drug with a short $t_{1/2}$ (**lorazepam, oxazepam, alprazolam**).
- **The substituted drug should be given for several wks** before gradual tapering begins.
- Adjunctive use of **imipramine, valproic acid, or buspirone** **can help to** reduce withdrawal symptoms during the BZ taper.

Drug interaction

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- The combination of **BZs with alcohol or other CNS depressants may be fatal.**
- **Alprazolam** dose should be reduced by 50% if **nefazodone or fluvoxamine** is added.

Dosing and administration

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- **Initiate with low doses**, and dosage adjustments can be made weekly
- Treatment of acute anxiety generally should **not exceed 4 wks.**
- BZs can be given as **PRN** basis
- BZs with a long $t_{1/2}$ may be dosed **QD** at bedtime

Treatment of Panic Disorder

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General Principles

- SSRIs, TCAs, and CBT are similarly effective.
- FDA approved:
 - ▣ Alprazolam, clonazepam, sertraline, paroxetine, and venlafaxine
- SSRIs are first-line agents, but BZs are most used
- Most pts without **agoraphobia** improve with drug alone, but
- if agoraphobia is present, CBT typically is initiated concurrently.

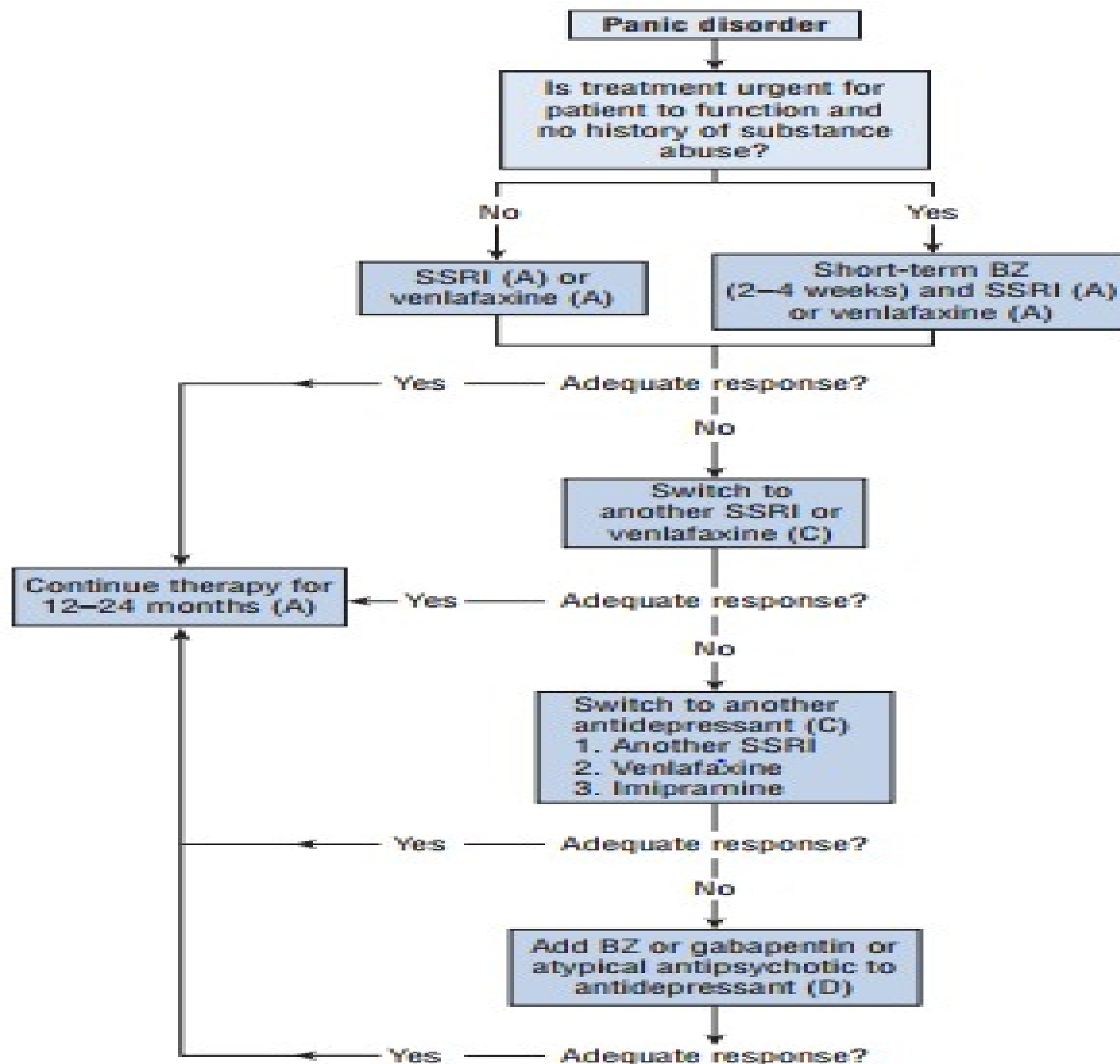


TABLE 66-12

Drugs Used in the Treatment of Panic Disorder

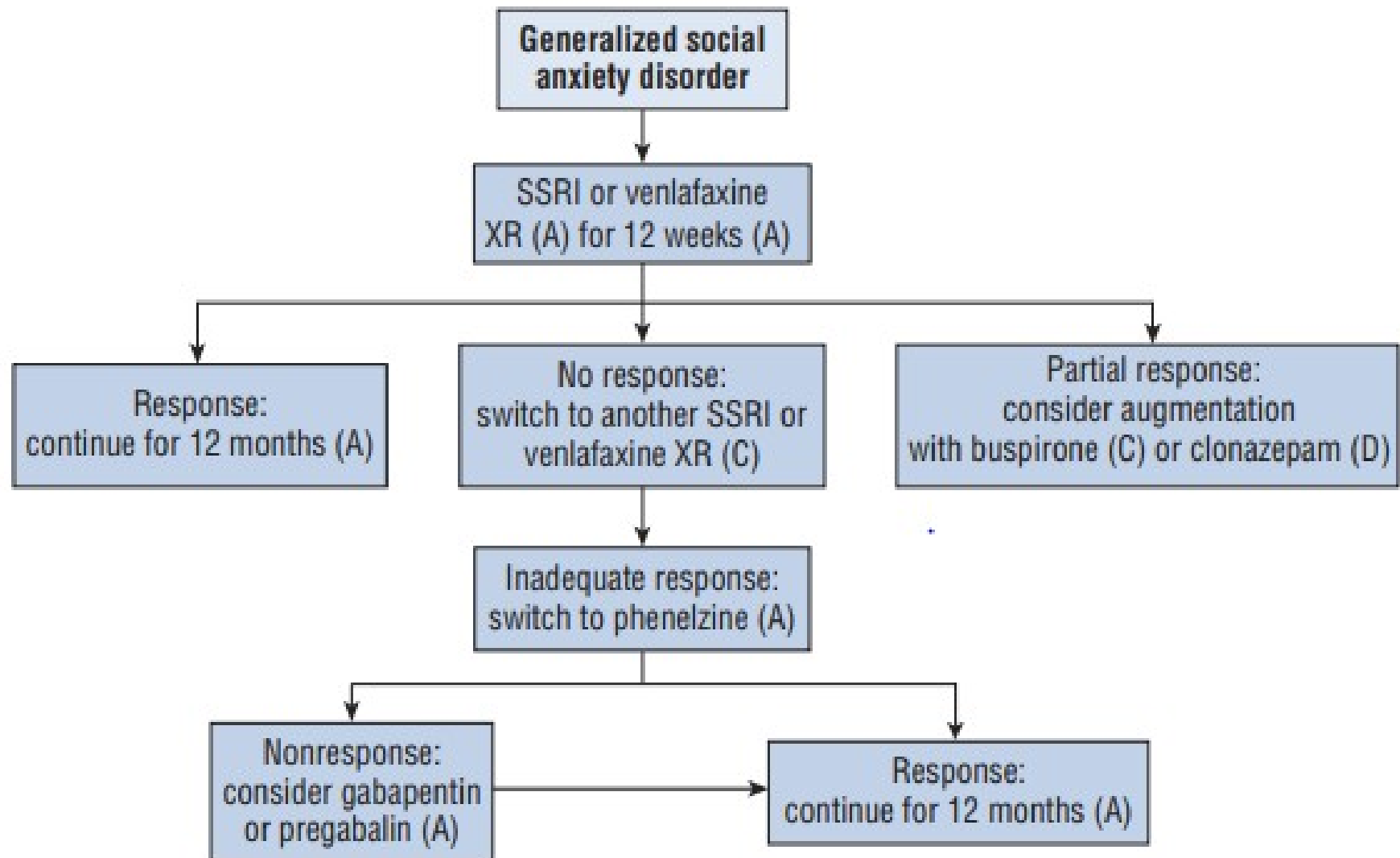
Class/Generic Name	Brand Name	Starting Dose	Antipanic Dosage Range (mg)	Comments
Selective Serotonin Reuptake Inhibitors				
Citalopram	Celexa	10 mg/day	20–40	Dosage used in clinical trials; maximum dose limited by QT prolongation; available generically
Escitalopram	Lexapro	5 mg/day	10–20	Dosage used in clinical trials; available generically
Fluoxetine	Prozac	5 mg/day	10–30	Available generically
Fluvoxamine	Luvox	25 mg/day	100–300	Available generically
Paroxetine	Paxil	10 mg/day	20–60	FDA-approved; available generically
	Pexeva Paxil CR	12.5 mg/day	25–75	
Sertraline	Zoloft	25 mg/day	50–200	FDA-approved; available generically
Serotonin–Norepinephrine Reuptake Inhibitor				
Venlafaxine XR	Effexor XR	37.5 mg/day	75–225	FDA-approved; available generically
Benzodiazepines				
Alprazolam	Xanax	0.25 mg three times a day	4–10	FDA-approved; available generically
	Xanax XR	0.5–1 mg/day	1–10	
Clonazepam	Klonopin	0.25 mg once or twice per day	1–4	FDA-approved; available generically
Diazepam	Valium	2–5 mg three times a day	5–20	Dosage used in clinical trials; available generically
Lorazepam	Ativan	0.5–1 mg three times a day	2–8	Dosage used in clinical trials; available generically
Tricyclic Antidepressant				
Imipramine	Tofranil	10 mg/day	75–250	Dosage used in clinical trials; available generically
Monoamine Oxidase Inhibitor				
Phenelzine	Nardil	15 mg/day	45–90	Dosage used in clinical trials

Social Anxiety Disorder

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- SAD pts often respond **more slowly** and **less completely**
 - At least 1 yr of maintenance therapy
- **Long term treatment** may be needed for pts with unresolved symptoms, comorbidity, an early onset of dx, or a prior history of relapse.

- **CBT** (exposure therapy, cognitive restructuring, relaxation training, and social skills training) and drugs are considered equally effective in SAD
- CBT can lead to a greater likelihood of maintaining response after treatment termination.
- CBT and social skills training are effective in **children** with SAD.



- SSRIs and venlafaxine can be used in children 6 to 17 yrs of age.
- But closely monitor for increased risk of **suicidality up to 24 yrs.**
- Approved first-line agents for generalized SAD
 - **Paroxetine, sertraline, and venlafaxine XR**
- With SSRI treatment, the **onset of effect** is delayed 4 to 8 wks

- **Fluoxetine and TCAs** are not effective for SAD.
 - Limited data suggest efficacy of citalopram.
- **BZs** should be reserved for
 - pts at low risk of substance abuse,
 - those who require rapid relief, or
 - those who have not responded to other therapies.

- **Gabapentin** was effective for SAD, and onset of effect was 2 to 4 wks.
- **β-Blockers** blunt the peripheral autonomic **symptoms of arousal** (rapid HR, sweating, blushing and tremor) in performance-related anxiety
- ✧ **For specific SAD**, 10 to 80 mg of **propranolol** or 25 to 100 mg of **atenolol** can be taken 1 hr before the performance.

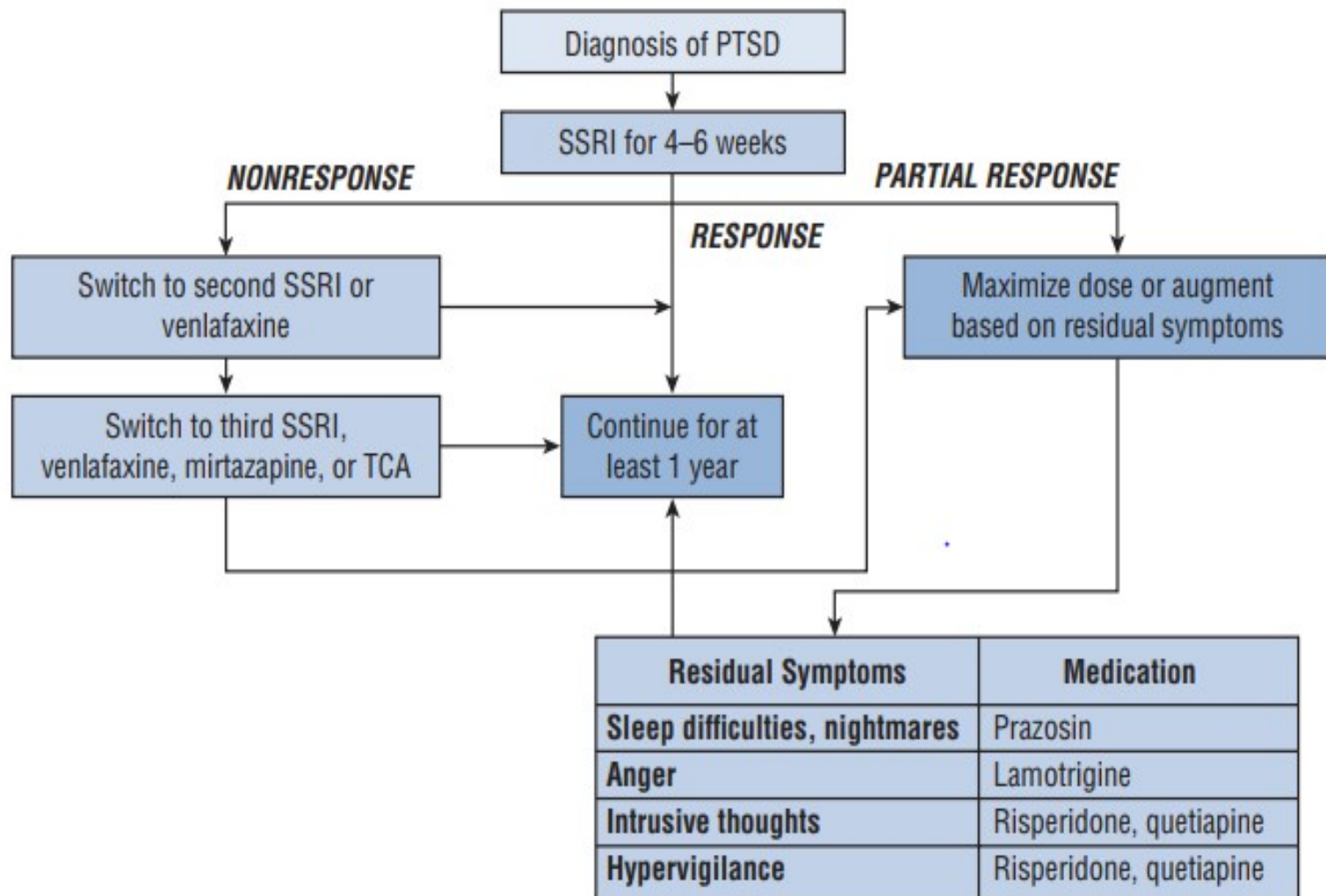
- **Augmentation:** Incomplete response to a first-line agent with **buspirone or clonazepam**
- **Phenelzine** is effective, but is reserved for treatment-resistant pts
- Pts with SAD should be monitored for symptom response, adverse effects, and overall functionality and QOL.
- Pts should be asked to keep a diary to record symptoms and their severity.

Posttraumatic Stress Disorder

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- Immediately after the trauma, pts should receive treatment individualized to their presenting symptoms (**non-BZ hypnotic, short courses** of CBT).
- Brief courses of CBT in close proximity to the trauma resulted in lower rates of PTSD.
- **The SSRIs are first-line**
- **Venlafaxine, TCAs and MAOIs** may also be effective, but they have less favorable side effect profiles.

- **Sertraline** and **paroxetine** are approved for acute treatment of PTSD
- **Sertraline** is approved for long-term management
- **Antiadrenergics** (prazosin) and **atypical antipsychotics** (risperidone, quetiapine, and olanzapine) can be used as **augmenting agents**.



- Pts should be monitored for
 - Symptom response,
 - Side effects and
 - Treatment adherence

THE END!!!!!!